

Problem

Design a problem to help other students better understand how to convolve two functions together.

Step-by-step solution

Step 1 of 1

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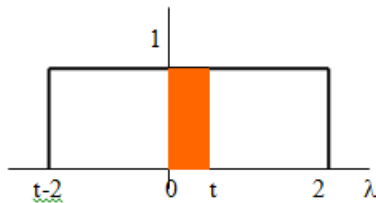
Although there are many ways to solve this problem, this is an example based on the same kind of problem asked in the third edition.

Problem

Suppose that $f(t) = u(t) - u(t-2)$. Determine $f(t) * f(t)$.

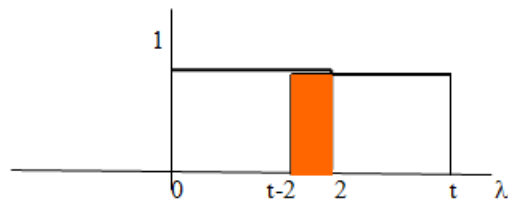
Solution

For $0 < t < 2$, the signals overlap as shown below.



$$y(t) = f(t) * f(t) = \int_0^t (1)(1) d\lambda = t$$

For $2 < t < 4$, they overlap as shown below.



$$y(t) = \int_{t-2}^2 (1)(1) d\lambda = t \Big|_{t-2}^2 = 4 - t$$

Thus,

$$y(t) = \begin{cases} t, & 0 < t < 2 \\ 4 - t, & 2 < t < 4 \\ 0, & \text{otherwise} \end{cases}$$